

AI Hardware Design League (AI-HDL) Webinar

2 Detailed Overview

Webinar Agenda Overview

This webinar, the second in a series, provides a deep dive into the AI-HDL competition structure and the initial challenge. The competition officially starts in **January**.

The webinar agenda covered the following topics:

1. Recap: The AI-HDL Vision & Structure
2. The Core Challenge: Project Scope & Base System
3. Methodology: AI-First & Dual-Toolchain
4. The 16-Week Journey: A Deep Dive
 - Phase 1: Base Design Expansion
 - Phase 2: Design Evaluation & PPA Optimization
 - Phase 3: Security Evaluation & Threat Mitigation
 - Phase 4: Netlist to Chip Tapeout
5. Next Steps & Q&A

Crucial Note for Registration: The current registration deadline for the competition is the **end of November**.

16-Week Competition Structure Recap

The competition is structured into four distinct phases, culminating in a physical chip tapeout:

- **Phase 1: Base Design Expansion (DP#1):** Focuses on adding new functionality to the base chip.
- **Phase 2: Design Evaluation & PPA Optimization (DP#2):** Focuses on improving the Power, Performance, and Area (PPA) efficiency of the design.
- **Phase 3: Security Evaluation & Threat Mitigation (DP#3):** Teams submit their designs, which will then be "hacked" by the organizing team to identify vulnerabilities. The teams must then fix the security flaws.
- **Phase 4: Netlist to Chip Tapeout (DP#4):** The final stage, where the Register-Transfer Level (RTL) code is converted to a GDSII file (the final physical layout file for manufacturing).

Key Technologies

The competition leverages a three-part technical foundation:

- **RISC-V:** An open-source Instruction Set Architecture (ISA) that teams will build upon. This provides a flexible and modifiable core.
- **AI (LLMs):** Large Language Models are the primary tool used for **design generation**.
- **EDA Tools (Electronic Design Automation):** Teams will gain hands-on experience with both open-source and commercial tools for design implementation and verification.

Methodology: AI-First

The core methodology for the competition is **AI-First**:

- **Key Rule:** The core hardware design must be generated via prompting LLMs.
- **Scoring:** Is based on the effective utilization of LLMs for design and optimization.
- **Minor manual corrections are allowed** to the LLM-generated design.
- **Documentation:** Participants must submit their **prompt logs** as a crucial part of the documentation to demonstrate their use of the AI tools.
- **Support:** Dedicated mentors and industry experts will be available via **Discord** (QR code provided in the webinar) to help teams with technical questions and overcome any challenges, especially regarding bugs or fixing issues in the AI-generated code.

The Core Challenge: Customizing a RISC-V Core

The project is a **Core-Based Implementation**. Teams will not be building a CPU from scratch.

- **You will be provided** with a functional, base RISC-V processor core.
- **Your primary task** is to extend this core with new, custom functionality (conceptually, adding a "guest room" to the main "house").
- **Focus** is on modifying and optimizing the provided core, not rebuilding the complex CPU logic.

The "Guest Room" Specifications

Teams will integrate a custom module into the RISC-V core using a full peripheral template (a guest room connected to the main hallway).

- **Data Size:** 32-bit (4 Bytes), the full width of the CPU.
- **Complexity:** High (the final module should handle complex operations).
- **Use Cases (Examples):** AI Accelerator (Matrix Math), Encryption Engine, Math Coprocessor (Faster calculations).

Design Phase #1 (DP#1): Base Design Expansion

(January)

The objective of the first phase is to integrate the custom module into the RISC-V core and ensure it is fully functional.

Timeline	Milestone/Activity
January 15	League Starts with DP#1: Base Design Expansion
January 29	MR#1 (Milestone Review) & OH#1 (Office Hours)

Objective: Integrate a custom module (e.g., accelerator, or instruction extension) into the base RISC-V core.

Deliverable: Functionality, Verification, and Documentation. (Initial speed and size are **not** the primary focus here).

DP#1: Key Concepts - RTL & Synthesis

- **RTL (Register-Transfer Level):** A high-level, behavioral description of your chip (your Verilog code).
- **Synthesis:** The process of converting your RTL into a structural description (a "netlist") made of standard logic gates (NAND, NOR, etc.).
- **Your Objective:** Produce an RTL code that is **synthesizable** (no errors) and **functionally correct**.

DP#1: Weekly Goals

Week	Date	Goal
Week 1	Jan 15-21	Project Kickoff & Base Integration: Develop your project plan and block diagrams for your new module. Use LLMs to generate Verilog for your new module and a simple, standalone testbench. Connect your new module to the top-level core.
Week 2	Jan 22-28	Full-Core Simulation, Verification & Milestone Prep: Run the full simulation suite to ensure you didn't break anything. Run a baseline synthesis to get initial PPA estimates. Fix any bugs, clean up your code, reports, AI prompt logs, and tag your final commit on GitHub .

DP#1: Key Tools

Category	Tool	Description
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Category	Tool	Description
Simulation (Verification)	Verilator	Open-source simulator that converts your Verilog to C++ for high-speed simulation.
	GTKWaves	The waveform viewer you will use to visually debug your design.
Synthesis (Implementation)	Yosys	The open-source synthesis tool you will use to check for synthesis errors and get a baseline report.
	Cadence Genus	The commercial alternative you will also have access to.

DP#1: Final Deliverables

1. **Synthesizable RTL:** Your complete, modified RISC-V core Verilog code.
2. **Verification Results:** All testbenches (provided and custom) must pass.
3. **Baseline Synthesis Report:** An initial report from Yosys/Genus showing area and timing estimates.
4. **DP1 Report & Documentation (required):**

◦ Full RTL module package.

◦ **LLM prompt logs (Crucial!).**

◦ Write-up of your Design choice, PPA results, and schematic.
5. **GitHub Tag:** A final commit tagged as "**DP1-Submission**".

DP#1: Scoring Rubric

Category	Weight	Criteria
Module Implementation	30%	Completeness & correctness of your new module.
Verification & Testing	25%	All testbenches pass; quality of custom unit tests.
Documentation & Planning	25%	Clear project plan, block diagrams, and LLM prompt logs.
Code Quality & Repository	20%	Use of GitHub tags, clean commits, code comments.

Design Phase #2 (DP#2): Design Evaluation & PPA Optimization (February)

The second phase is dedicated to analyzing and optimizing the design generated in DP#1 for efficiency.

Timeline	Milestone/Activity
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Timeline	Milestone/Activity
February 12	DP#2: Design Evaluation and PPA Optimization Starts
February 26	MR#2 (Milestone Review) and OH#2 (Office Hours)

Objective: Take your functional DP#1 design and make it efficient. Optimize for **Power, Performance, and Area (PPA)**.

Deliverable: Analysis, Microarchitectural Tweaks, and Trade-offs.

DP#2: Key Concepts (What is PPA?)

- **Power:** How much power your chip consumes (measured in watts). **Lower is better.**
- **Performance:** How fast your chip runs (measured in frequency, e.g., MHz, or delay, e.g., nanoseconds). **Higher frequency / Lower delay is better.**
- **Area:** How much physical silicon space your chip uses. **Smaller is better.**
- **The "Iron Triangle":** Improving one (e.g., Performance) could impact others (e.g., Power, Area). Your objective is to find the best trade-off.

DP#2: Key Concepts (What is STA & CG?)

- **Static Timing Analysis (STA):** The process of analyzing a design's timing **without simulation**. It finds the critical path, the slowest path in your design, which limits your max clock speed.
- **Clock Gating (CG):** A popular power-saving technique. The idea: if a part of your chip isn't being used, stop sending it the clock signal. **No clock = no switching = no power consumption.**

DP#2: Weekly Goals

Week	Date	Goal
Week 1	Feb 12-18	Baseline Analysis & Planning: Run a full PPA analysis on your final DP#1 design to get a baseline. Create an "Optimization Plan" listing your PPA goals. Implement microarchitectural tweaks to fix your critical path (e.g., add pipeline stages). Implement power-saving features like clock gating.
Week 2	Feb 19-25	Performance Optimization & Submission: Run regression tests to ensure new optimization tweaks did not break normal functionality. Analyze PPA trade-offs (e.g., did power saving hurt timing?). Generate final PPA reports. Create a "before-and-after" comparison report to show improvements. Tag your DP#2 submission on GitHub.

DP#2: Final Deliverables

The successful completion of Design Phase #2 requires the following final submission components:

1. **Optimized RTL:** Your modified, functionally correct, and PPA-optimized Verilog code.
2. **Final STA & PPA Reports:** The final sign-off reports showing your new and improved Power, Performance, and Area (PPA) numbers, as well as the Static Timing Analysis (STA) results.
3. **Optimization Report:** A document containing:
 - Your baseline PPA analysis from Week 1.
 - A "before-and-after" comparison highlighting your final improvements.
 - A write-up justifying the design trade-offs made during the optimization process (e.g., accepting slightly lower frequency for a significant reduction in power consumption).
4. **GitHub Tag:** A final commit tagged as "**DP2-Submission.**"

DP#2: Scoring Rubric

Category	Weight	Criteria
Performance Optimization	25%	Measurable improvements in timing/frequency metrics.
Power Optimization	25%	Implementation of power-saving features (e.g., clock gating) and a measurable power reduction.
Final Results & Reporting	20%	Quality of before/after comparisons and ensuring all regression tests are clean (all pass).
Trade-off Justification	15%	Clear documentation of <i>why</i> certain PPA trade-offs were made.
Baseline PPA Analysis	15%	Thorough analysis of the initial DP#1 design and clear identification of bottlenecks.

Design Phase #3 (DP#3): Security Evaluation & Threat Mitigation

DP#3: Overview

- **Timeline:** Starts **March 12**
- **Milestone Review & Office Hours (MR#3 & OH#3):** March 26
- **Objective:** Assess your PPA-optimized design for hardware vulnerabilities and implement countermeasures to make it secure. This phase is often the most "eye-opening" for students, as secure chip design is a critical and often overlooked area in curriculum.
- **Deliverable:** Security Assessment, Countermeasure Implementation, and Validation.

DP#3: Key Concepts (What is Hardware Security?)

- **Hardware Vulnerability:** A flaw in your design (a weakness in the logic itself) that an attacker can exploit.
 - *Critical Distinction:* Unlike software, **hardware bugs cannot be addressed post-fabrication** (after the chip is printed).
- **Common Weakness Enumeration (CWE):** A dictionary of known hardware and software weakness types.
 - *Examples:* Side-channels, Hardware Trojans, Debug misuse, and Buffer overflows.
- **Your Objective:** You will act as a security audit team. You will find potential CWEs in your design, implement fixes, and use the AI tools (LLMs) to help verify your design for vulnerabilities.

DP#3: Weekly Goals

Week	Date	Goal
Week 1 (Mar 12-18):	Security Assessment & Implementing Security Features	(1) Create a "Security Review Document" listing potential CWEs in your DP#2 design. (2) Develop a "Mitigation Plan" for the features you will add. (3) Implement at least one major security countermeasure (e.g., Add privilege-level checks, data input authentication, or logic to randomize power consumption).
Week 2 (Mar 19-25):	Validation, Hardening & Documentation	(1) Test your security feature (how do you prove it works?). (2) Run all regression tests to ensure you didn't break normal functionality. (3) Analyze the PPA impact (i.e., how much did your security feature "cost" in area or timing?). (4) Write your final DP#3 Security Report and tag your secure RTL for submission.

DP#3: Final Deliverables

1. **Secure RTL:** Your modified, hardened, and functionally correct Verilog code.
2. **Security Evaluation Report:** A comprehensive document containing:
 - Your vulnerability assessment from Week 1.
 - A detailed explanation of the countermeasures you implemented.
 - Validation results (how you proved the fix works).
 - PPA Overhead Analysis (the "cost" of your security feature).
3. **GitHub Tag:** A final commit tagged as "DP3-Submission."

DP#3: Scoring Rubric

Category	Weight	Criteria
Security Feature Implementation	30%	Meaningful, well-implemented countermeasures.
Validation & Hardening	25%	Comprehensive security testing to prove the fix works.

Category	Weight	Criteria
Security Assessment	20%	Thorough vulnerability review and a well-structured plan.
Documentation & Reporting	15%	Clear DP#3 report, detailed explanation of mitigations & trade-offs.
Design Stability	10%	Functional correctness is preserved; all regression tests are clean.

Design Phase #4 (DP#4): Netlist to Chip Tapeout (April)

DP#4: Overview (For Finalists Only)

- **Timeline: April 9 – April 22** (The competition phase).
- **Milestone Review & Office Hours (MR#4 & OH#4):** April 23.
- **Objective:** Take your secure, optimized RTL design and turn it into a final, manufacturable physical layout (GDSII file).
- **Objective:** From RTL to Silicon.
- **Deliverable:** Physical Design, "Back-End" Flow, and Sign-off.

DP#4: Key Concepts (The "Back-End" Flow)

This phase represents the final steps to transition from logical code to a physical chip layout:

Step	Process	Description
1. Floorplanning	Defining the chip's "floorplan" where the major functional blocks (CPU, Memory, I/O) along with the logic blocks, I/O pins, and power grid will go.	
2. Place & Route (P&R)	Placement: The tool places all the millions of standard cells onto the floorplan. Routing: The tool connects all the cells with tiny metal wires (traces).	
3. Sign-off (DRC/LVS)	DRC (Design Rule Check): Ensures your layout is manufacturable (e.g., no wires are too close or too thin). LVS (Layout vs. Schematic): Ensures the physical layout matches the logical circuit/design (RTL logic) you intended.	

DP#4: Weekly Goals

Week	Date	Goal
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Week	Date	Goal
Week 1 (Apr 9-15):	Synthesis, Floorplanning, Placement, CTS (Clock Tree Synthesis) & Routing	(1) Run your final synthesis on the DP#3 design. (2) Create your chip's floorplan, including power grid and I/O pin placement. (3) Run the main Placement, CTS, and Routing tools. (This is a highly iterative process where you will run the tool, analyze timing, tweak settings, and run it again until timing closure is met).
Week 2 (Apr 16-22):	Sign-off Verification & Final Tape-out Package	(1) Run your final STA, DRC, and LVS checks. (2) Your goal is "DRC/LVS Clean" (zero errors). This is the bar for a manufacturable chip. (3) Package your final GDSII file. (4) Write your final, comprehensive project report covering the entire 16-week journey (DP#1-DP#4).

DP#4: Final Deliverables

- Final GDSII File:** The manufacturable layout of your chip (the geometric data file for fabrication).
- Sign-off Reports:** Clean DRC, LVS, and final STA reports (showing zero errors).
- Final Project Report:** A comprehensive report detailing your entire journey from DP#1 to DP#4, including your final PPA metrics, security features, and reflections on the process.
- GitHub Tag:** A final commit tagged as **"DP4-Submission."**

DP#4: Scoring Rubric

Category	Weight	Criteria
Physical Design Flow Execution	30%	Successful, complete flow (Synth, P&R, CTS); no critical gaps.
Sign-off Quality	20%	DRC/LVS clean reports (A manufacturable design).
Timing Closure & PPA	20%	Final STA reports are clean; final PPA metrics are well-documented.
Submission Package	20%	Complete GDSII, netlist, constraints, and final comprehensive report.
Presentation & Clarity	10%	Clear repository organization and project story.

Final Competition Summary

How are the Winners Chosen?

Final ranking is based on a **Cumulative Weighted Score** across all four design phases (Total: 100%):

Design Phase	Focus	Weight
DP#1	Functionality	20%
DP#2	PPA Optimization	25%
DP#3	Security	25%
DP#4	Layout	30%

- The **Top 3 Teams** with the highest cumulative score will be recognized as Winners.
- The **Top-Performing Team** will win a **Chip Tapeout** (having their design manufactured as a physical chip).

Bonus Opportunities & Penalties

Category	Examples
Bonus Points (For Exceptional Innovation)	Exceptional, creative use of AI/LLMs. Implementing an advanced feature. Contributing a bug-fix or feature back to an open-source tool.
Major Penalties (Major Deductions)	A design that doesn't simulate or synthesize at final submission. Failing to submit a GDSII file in DP#4. Failing DRC/LVS checks (not a manufacturable design). Incomplete documentation (e.g., missing AI prompt logs).

Your Journey Starts Here: Next Steps

Action	Details
Registration Closes	November 30th, 2025. All team members must register individually using their institutional email address.
Attend Next Webinar	Initial RTL Demo & Environment Walkthrough (Dec 18). A live demo of the base core and the open-source tool setup will be provided.
Join the Discord	Start talking with other participants and mentors now.

Note: All competition materials, including video tutorials and the base design core, will be made available through the competition website and Discord channel.

Q&A: Competition Logistics & Enrollment

Team Composition and Submission Process

- **Individual Registration:** Every participant in the team must register individually on the website using their institutional email.
- **Team Size Requirement:** Teams must consist of **3 to 5 members**. If a team registers with only two members, the organizers will assign a third member from the *same institution* to their team to ensure they meet the minimum requirement.
- **Team Lead & Communication:** Once a team is formed, **one person will be designated as the Team Lead**.
 - This individual will be the **main point of contact** for the organizers.
 - The Team Lead is responsible for all **GitHub submissions** for the team.
 - The Team Lead acts as the central link to receive resources and pose any questions (including tool or technical troubleshooting) to the dedicated mentors.

Official Resources and Timeline

- **Competition Timeline:** The entire competition timeline, including all Design Phase (DP) and Milestone Review (MR) dates, is available on the **AI-HDL website** under the "Timeline" tab.
- **Webinar Recordings:** The recordings for all past webinars, including this one, will be posted on the AI-HDL website. The registration link for a webinar will be updated to a link for the YouTube video recording shortly after the session.
- **Next Key Webinar:** Participants are advised to attend **Webinar 3 (Dec 18)**, which will feature a live demo of the base RISC-V core design and a walkthrough of the tool environment setup.

Q&A: Technical & Development Process

Educational Resources and Prerequisites

- **Learning Materials:** The website contains an extensive section of "**Educational Resources**" for participants who may not have a deep background in all areas of hardware design.
- **Content Includes:** Training materials on:
 - The fundamentals of hardware and semiconductors (e.g., what is an FPGA, ASIC, CPU architecture).
 - An introduction to **Verilog** (the hardware description language used in the competition).
 - How to utilize AI for hardware design, including using LLMs for ASIC design.

Asynchronous Workload & Flexibility

- **Self-Paced Work:** The AI-HDL competition is **asynchronous**. Teams are encouraged to work on their own time and schedule to best accommodate their academic commitments (e.g., final

exams in January).

- **Design Phase Submission:** While the timeline suggests completing the phases sequentially, teams can **submit all four Design Phases at once** if they wish. The overall cumulative score at the end is what matters. However, following the suggested schedule is highly recommended for pacing and iterative improvement.
- **Suggested Weekly Commitment:** The organizers suggest a weekly commitment of approximately **6 hours per team**, broken down as:
 - 1 hour for weekly check-ins/meetings (Milestone Reviews, Office Hours).
 - 5 hours for project work, which can be distributed among team members.

Advanced Features and Bonus Points

- **Advanced Feature Definition (DP#1):** The core task involves creating a new, custom module (a "guest room") for the provided RISC-V core. An **"advanced feature"** refers to the complexity and novelty of the function built into this module, such as a specialized **AI Accelerator, an Encryption Engine, or a Math Coprocessor**.
- **Bonus Points Opportunities:** Teams can earn bonus points for:
 - **Exceptional use of AI/LLMs** in the design process.
 - Implementing highly **advanced features or optimizations** in DP#1 or DP#2.
 - Identifying and reporting a **novel security vulnerability (CWE)** in the design (DP#3) that the organizing team had not previously found.
- **DP#4 Back-End Tooling:** For Design Phase 4 (Netlist to Chip Tapeout), teams are given flexibility in how they manage the complex transition between the synthesis tools (Yosys/Cadence) and the physical design tools (OpenROAD/Synopsys). Creating a **unified script or makefile** to automate the process of managing intermediate files, data passing, and constraints would be considered a valuable contribution to the overall design process and scored accordingly.

Final Review: Key Takeaways & Next Steps

- **Overall Goal:** The competition's main goal is for every participant to **learn a valuable, in-demand hardware design skill set** using modern AI tools and gain experience with industry-standard flows.
- **Winner Recognition:** The competition will recognize:
 - The **Top 3 teams** in both the Undergraduate and Graduate divisions based on the cumulative weighted score across all four Design Phases.
 - The overall single best design from either division will win the prize of having their chip **fabricated (Chip Tapeout)**.
 - All participants who actively engage in the competition will receive a **certificate of completion**.
- **Critical Next Steps:**
 1. **Register:** Finalize your team and **register by November 30th, 2025**.

2. **Join Discord:** The Discord server is the official hub for all communication, peer-to-peer discussions, and mentor support. (Use the QR code provided in the webinar).
3. **Attend Webinar 3:** Mark your calendar for **December 18th** for the initial RTL demo and environment walkthrough.